

Innovation & User-Centered Design Addendum

How the project goes beyond prediction and turns machine-learning insight into a practical digital-wellness experience.

Core idea: the ML model is the analytical engine, but the product closes the loop by helping the user understand the result, reflect on their day, take a realistic action, and return to track progress over time.

1. From One-Time Prediction to a Longitudinal Wellness Companion

The project does not stop after a single risk prediction. Each user can create a local account, return later, restore previous information, and view 7-day, 30-day, and 90-day progress. Missing check-in days are preserved as missing data rather than being treated as zero usage, which keeps trend summaries more honest. This turns the application from a one-time classifier into an ongoing behavior-reflection tool.

2. Explainable AI That Leads to Action

The system combines calibrated model output, SHAP explanations, dataset-relative percentiles, and user context. Instead of only saying “reduce screen time,” it identifies which behaviors influenced the current result and suggests practical replacement activities based on available time, goals, and preferences. The goal is to make the prediction understandable and usable, not simply accurate.

3. Mood Reflection and Daily Note

Digital wellness is not only about hours on a screen. The application allows users to record how they feel, rate stress, happiness, focus, and related daily factors, and write a short personal note. The note is designed as a private reflection space: users can put the day into words, review it later, and connect subjective wellbeing with their own usage history. These entries are not used to claim medical diagnosis or causation.

4. Focus Sound Space: A Practical Support Feature

A distinctive product feature is the local Focus Sound Space. Users can choose from white, pink, and brown noise, as well as separate nature-style soundscapes such as rain, streams, birds, and waterfall sounds. They can play one sound or intentionally mix multiple layers, control each layer’s volume, set a timer, or use an “Always” mode until they pause or stop playback.

- **Independent volume control:** Sound volume can be adjusted inside the app without changing the device’s overall audio level.
- **Mix Studio:** Users can combine sound layers instead of being limited to one fixed recording.
- **Sleep / focus timer:** Playback can stop automatically after a chosen duration.

- **Local and optional:** The feature does not require a cloud service and is not presented as treatment for ADHD, anxiety, or any medical condition.

5. Human-Centered Interface, Not Decorative UI

The interface was designed to reduce friction and make the system comfortable to revisit.

Readable typography, day/night modes, calm visual themes, persistent accounts, progress views, and local-first storage are product decisions that support continued use. The interface is therefore not the innovation by itself; it is the delivery layer that makes the machine-learning analysis understandable and approachable.

6. What Differentiates This Project

Typical analyzer	One-time input → prediction → generic advice
Digital Wellness Analyzer	Prediction → calibrated score → SHAP explanation → personalized action → reflection → saved history → progress tracking → optional focus support

Final message for judges: The project's contribution is the combination of technically careful machine learning with a user-centered follow-up loop. It aims to make digital-wellness analysis something a person can understand, act on, and return to - not just a score displayed once.

Scope note: This is a wellness-support and behavior-reflection system, not a medical or clinical diagnostic tool. Sound features are optional support tools and are not presented as treatments.